

Strip cropping in organic agriculture results in 15% higher ground beetle richness and 30% higher activity density than monocultures

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This study presents **important** findings on increased ground beetle diversity in strip cropping compared with crop monocultures. **Solid** methods are used to analyze data from multiple sites with heterogeneous systems of mixed crops, allowing broad conclusions, albeit at the expense of lacking taxonomic specificity. The work will be of interest to all those applying plant diversity treatments to improve the diversity of associated animals in agricultural fields.

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Abstract

Global biodiversity is declining at an unprecedented rate, with agriculture as one of the major drivers. There is mounting evidence that intercropping can increase insect biodiversity while maintaining or increasing yield. Yet, intercropping is often considered impractical for mechanized farming systems. Strip cropping is a type of intercropping that is compatible with standard farm machinery and has been pioneered by Dutch farmers since 2014. Here, we present ground beetle data from four organically managed experimental farms across four years. Ground beetles are sensitive to changes in habitats and disturbances, and hold keystone positions in agroecosystem food webs. We show that strip cropping systems can enhance ground beetle biodiversity, while previous research shows that these increases have been achieved without incurring major yield loss. Strip cropped fields had on average 15% more ground beetle species and 30% more individuals than monocultural fields. The higher ground beetle richness in strip crops was explained by the merger of crop-related ground beetle communities, rather than by ground beetle species unique to strip cropping systems. The increase in field-level beetle species richness in organic agriculture through strip cropping approached increases found for other readily deployed biodiversity conservation methods, like shifting from conventional to organic agriculture (+19% - +23%). This indicates

that strip cropping is a potentially useful tool supporting ground beetle biodiversity in agricultural fields without compromising food production.

Introduction

Insects account for 80% of the animal species in the world and, therefore, the recently reported unprecedented rate of insect decline is cause for alarm about the state of biodiversity on Earth (Dirzo et al., 2014 [↗](#); Hallmann et al., 2017 [↗](#); Van Klink et al., 2020 [↗](#)). An array of biodiversity metrics provide strong evidence for declines of especially terrestrial insects across all continents (Van Klink et al., 2020 [↗](#)). This includes declines in abundance of both common and rare species (Hallmann et al., 2017 [↗](#); Pilotto et al., 2020 [↗](#); Seibold et al., 2019 [↗](#); van Klink et al., 2024 [↗](#)) and in total species richness and changes in assembly composition (Blowes et al., 2022 [↗](#); Wagner et al., 2021 [↗](#)). Insects are essential for crop production through their role in decomposition, pollination, pest control and sustaining food webs. Therefore, erosion of insect communities can have potentially devastating effects on ecosystem functioning, provision of ecosystem services and ultimately on human civilization (Dirzo et al., 2014 [↗](#)). The main drivers of insect biodiversity decline are habitat loss due to conversion to agriculture, pollution, invasive species and climate change (Díaz et al., 2019 [↗](#); Müller et al., 2023; Wagner et al., 2021 [↗](#)). Strategies for biodiversity conservation in conjunction with adequate food production require understanding of how biodiversity responds to agricultural management (Cozim-Melges et al., 2024 [↗](#); Mupepele et al., 2021 [↗](#); Saunders, 2020 [↗](#)). Ideally, sustainable agricultural practices retain yield and enhance biodiversity, preventing the need to convert natural habitats to agricultural land to maintain food production (Tscharntke et al., 2021 [↗](#)).

Increasing crop heterogeneity can facilitate biodiversity conservation in highly productive agricultural landscapes without compromising yield (Martin-Guay et al., 2018 [↗](#); Sirami et al., 2019 [↗](#)). Crop diversification can enhance niche complementarity by creating heterogeneous habitats and increasing availability and diversity of resources (Lichtenberg et al., 2017 [↗](#); Tamburini et al., 2020 [↗](#)). A promising crop diversification strategy is strip cropping, where crops are grown in alternating strips, wide enough for using standard agricultural machines yet narrow enough to facilitate ecological interactions among crops (Ditzler et al., 2021 [↗](#)). Crops that are grown in strips may benefit from increased resource use efficiency, and the suppression of pests and diseases (Croijmans et al., 2024 [↗](#); Karssemeijer et al., 2024 [↗](#); Rakotomalala et al., 2023 [↗](#)) without major yield compromises (Campanelli et al., 2023 [↗](#); Juventia and van Apeldoorn, 2024 [↗](#); Van Oort et al., 2020 [↗](#)). Growing multiple crops on a field may foster a larger diversity of organisms than monocultures through greater plant species richness that cascades into richer herbivore and predator communities (Crutsinger et al., 2006 [↗](#); Cuperus et al., 2023 [↗](#)). Moreover, the expected increase in available and potentially complementary niches within the agricultural field due to higher spatial diversity in crops can result in admixture of communities related to individual crops (Hummel et al., 2012 [↗](#)), or the creation of completely new communities by enhanced richness of agriculture-related species, and/or the occurrence of species rarely found in agricultural fields (Risichen et al., 2021 [↗](#)). So far, it is not well understood if and how insect communities respond to strip cropping across distinct crops.

Here, we present data of four years of pitfall trapping of ground beetles at several moments during the growing season in 14 crops at four organic experimental farms across the Netherlands where strip cropping was compared to monocultures. Ground beetle communities are sensitive to changes in farming practices and are frequently used to examine agricultural sustainability (Holland and Luff, 2000 [↗](#); Makwela et al., 2023 [↗](#); Turin, 2022 [↗](#)). Furthermore, ground beetle species are important for maintaining ecological functions as they comprise scavengers and predators of (weed) seeds, detritivores (e.g., collembolas and earthworms) and herbivores (e.g., aphids and caterpillars). We first examine whether strip cropping fields have greater ground beetle richness and activity density than monocultural fields. Next, we present biodiversity metric

changes for individual crops to assess the benefits of strip cropping for ground beetle biodiversity. We then evaluate whether ground beetle community changes are caused by admixture of communities, whether these assemblages promote species associated with agricultural or with natural ecosystems, and whether they contain rare species. Lastly, we test for 12 common ground beetle genera whether their activity density is higher in strip cropping compared to monocultures.

Results

A total of 48,108 ground beetles belonging to 71 species were caught using pitfall traps over four years at four different organically managed experimental farms in The Netherlands: 40,153 at Almere; 3,777 at Lelystad; 1,126 at Valthermond; and 3,052 at Wageningen (Table S1, S2).

Strip cropping enhances ground beetle richness

Strip cropping fields had on average 15% higher ground beetle taxonomic richness than monoculture fields after rarefaction to the number of samples of the least-sampled crop configuration ($\beta_0 = 0.151$, SE = 0.044, $p < 0.001$; **Fig. 1c,d**). However, strip cropping fields did not harbour more species than monocultural fields with the highest ground beetle richness ($\beta_0 = -0.008$, SE = 0.037, $p = 0.821$; **Fig. 1d**). The difference in field-level taxonomic richness could not be explained by an increase in the number of ground beetle species per crop in strip cropping compared to monocultures. At crop-level, the 5% increase in ground beetle taxonomic richness in strip cropping was not statistically significant (**Fig. 1f**). Similarly, crop-level absolute evenness, inverse Simpson index and Shannon entropy did not differ significantly among crop configurations (**Fig. 1h-j**, S1c,d, S2). The effect of crop configuration on crop-level taxonomic richness was variable and was not associated with location or crop species. The effect ranged from 56% more species in potato in monoculture at Wageningen in 2022, to 136% more species in barley in strip cropping at Wageningen in 2020 (**Fig. 1k**).

Strip cropping enhances ground beetle activity density

Ground beetle activity density was on average 30% higher in strip cropping fields than in monoculture fields ($\beta_0 = 0.303$, SE = 0.121, $p = 0.012$; **Fig. 1e**), based on rarefaction. However, there was no significant difference in activity density between the strip cropping fields and monocultures that harboured the richest beetle communities ($\beta_0 = -0.110$, SE = 0.088, $p = 0.215$). Crop-level activity density of ground beetles was not affected by crop configuration (**Fig. 1g**, S1a,b).

Crop configuration alters ground beetle community composition

Ground beetle communities were significantly influenced by crop configuration, but the effects were highly dependent on the specific context created by the combination of location, year and crop (Fig. S3-S8, **Table 1**, S3). The context-dependency of configuration effects on ground beetle communities is illustrated in redundancy analyses of ground beetle assemblages per crop combination at Wageningen in 2021 and 2022 (Fig. S9). Here, we found distinct ground beetle communities among crop configurations for pumpkin, barley and potato in 2021, and for cabbage and oat in 2022. In the other cases the difference between crop configurations was not significant (Fig. S9, Table S4). Moreover, in all crop combinations except for potato-grass in 2021, the difference in ground beetle communities between the monocultures of the constituent crops were significant, while this was never the case for ground beetle communities of crops in strip cropping (Table S4). This indicates that strip cropping might lead to overlapping crop-related communities. However, these results could be spatially autocorrelated as samples from different crops were in closer proximity of each other in strip cropping than among monocultures (Fig S10d).

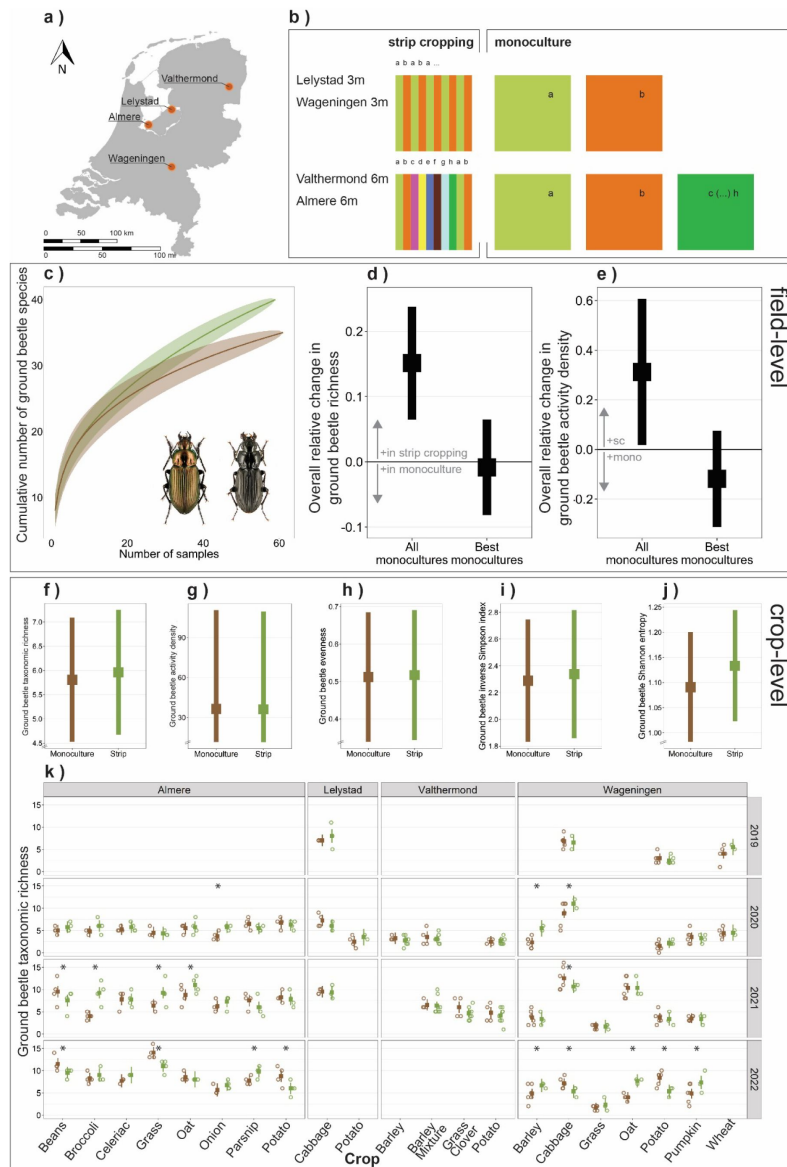


Fig. 1.

Effect of crop configuration (monoculture versus strip cropping) on ground beetle biodiversity.

(a) Location of experimental sites in the Netherlands. **(b)** Field set-up of the two crop configurations: monoculture and strip cropping. At Lelystad and Wageningen, strip cropping consisted of 3-m wide crop strips of two crops (pairs), and multiple crop pairs were assessed. At Almere and Valthiermond, strip cropping consisted of 6-m wide crop strips of eight crops combined (Fig. S10). **(c)** Sample-based species accumulation curves of all year series from monocultures (brown) and strip cropping (green), in Almere from 2021 and 2022. This was the only location where an equal number of samples were taken in the monocultures and in strip cropping on a similar area. Ground beetle species include *Poecilus cupreus* (left) and *Pterostichus melanarius* (right). Photo credit: Ortwin Bleich, retrieved from: www.eurocarabidae.de. **(d-e)** Overall relative change in field-level ground beetle **(d)** taxonomic richness and **(e)** activity density. Positive values indicate higher richness or activity density in strip cropping, negative values in monocultures. **(f-j)** Overall relative effect of crop configuration on ground beetle **(f)** taxonomic richness, **(g)** activity density, **(h)** absolute evenness, **(i)** inverse Simpson index, and **(j)** Shannon entropy. **(k)** Effect of crop configuration on ground beetle taxonomic richness for each combination of location, year and crop. Barley-mixture consists of a mixture of barley-bean (2020) or barley-pea (2021). Squares indicate estimated means, the bar indicates the 95% confidence interval. Asterisks indicate significant differences among the crop configurations. Empty panels indicate combinations of years and locations that were not sampled. When no estimated mean and confidence interval are shown, crops were not grown or sampled in that year. Open circles indicate individual year series to visualize sample size (Table S7).

Location	Predictor	Df	Sum sq	R2	F	P
All locations	Crop configuration	1	1.68	0.01	7.32	0.001
	Location	2	47.7	0.29	103.8	0.001
	Location : Year	6	10.7	0.06	7.72	0.001
	Location : Year : Crop species	31	33.3	0.20	4.67	0.001
	Crop configuration : Location	2	0.63	0.00	1.37	0.149
	Crop configuration : Location : Year	6	2.31	0.01	1.67	0.012
	Crop configuration : Location : Year : Crop species	31	14.8	0.09	2.08	0.001
	<i>Residual</i>	<i>245</i>	<i>56.3</i>	<i>0.34</i>		
	<i>Total</i>	<i>324</i>	<i>167.4</i>	<i>1.00</i>		

Table 1.

Effect of crop configuration on ground beetle community composition.

Results from permanova analysis using Hellinger's transformation for data from the three locations with species level data (see Table S3 for analyses per location). Crops were a nested variable within years, as these differed among years. Years were nested in locations, as the years that were studied differed among locations. Bold letters indicate significant effects ($\alpha = 0.05$).

Crop configuration alters abundance of common genera

We tested how crop configuration affected twelve common ground beetle genera (Table S5). We analysed this separately per location as some genera only occurred at specific locations. Four genera were more abundant in strip cropped fields in at least 1 location (*Anchomenus*, *Bembidion*, *Harpalus* and *Nebria*), whereas four genera were more abundant in monocultures in at least 1 location (*Amara*, *Calathus*, *Pterostichus* and *Trechus*) (Fig 2). The other four common genera (*Blemus*, *Clivina*, *Loricera* and *Poecilus*) were not significantly influenced by crop configuration. Furthermore, no ground beetle genus showed significantly contrasting responses to crop configuration among locations.

Crop configuration does not increase the number of rare ground beetle species

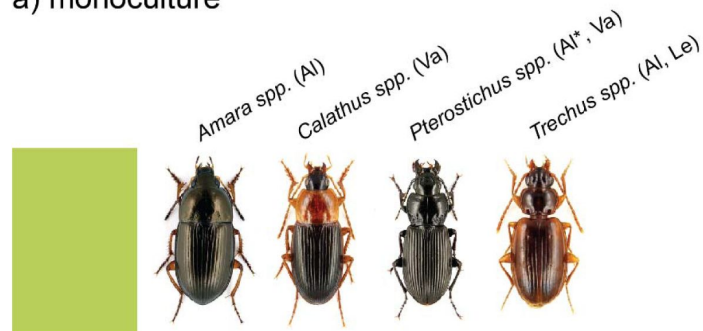
Among the 461 year series, we only found two rare species (according to waarneming.nl): one individual of *Microlestes minutulus*, and five individuals of *Harpalus signaticornis*. This latter species was found most often in a wheat monoculture, but this was likely due to a failed crop which created a very open habitat. *Harpalus signaticornis* is known to inhabit recently disturbed, dry, open habitats with limited vegetation (Turin et al., 2012), a situation similar to this sparsely-covered monoculture. All other species were common or relatively common species (Table S2). Furthermore, most ground beetle species were either ruderal habitat specialists or eurytopic species that occur in many different habitats (Table S2).

Discussion

It is well established that different crop types have distinct ground beetle communities (Eyre et al., 2009; Holland and Luff, 2000) and that increasing habitat diversity by including multiple crops in a field can enhance ground beetle diversity (Cuperus et al., 2024; Puliga et al., 2023). Our study shows that strip cropping increased field-level ground beetle richness by 15%. However, ground beetle communities within the same crop in a strip or in monoculture were mostly similar. This indicates that the 15% increase in richness at the field level can be mostly attributed by the higher number of crops in strip cropped fields that harboured crop-related ground beetle communities, and that there was only limited mixing of ground beetle species among crops in strip crops. This is in line with earlier findings that ground beetle movement is reduced by crop edges (Allema et al., 2014; Anderson et al., 2024). Further research on movement behaviours of ground beetles at crop edges might help explain how ground beetles distribute themselves within a strip cropping field, and whether they utilize the different resources provided by a more diverse cropping system.

The 30% higher ground beetle activity density in strip cropping fields compared to monoculture fields may be explained by a more stable and diverse habitat with refuges and alternative resources in strip crops (Ratnadass et al., 2012). Crop diversification enhances prey biomass for ground beetles (Lichtenberg et al., 2017) and increases weed seed richness as compared to monocultures (Ditzler et al., 2023), both of which reduce bottom-up control by increased food provision (Carbonne et al., 2022). Alternatively, the increase in activity density might be caused by higher movement of ground beetles, which in turn can be the consequence of food starvation (Wallin and Ekbom, 1994). Indeed, several papers show reduced abundances of herbivorous insects in strip- and intercropping (Alarcón Segura et al., 2022; Cuperus et al., 2023; Rakotomalala et al., 2023), which may be potential prey for ground beetles (Turin, 2000). Furthermore, strip cropping systems may support high herbivore reproduction in combination with a high predation pressure resulting in herbivore populations dominated by early life stages

a) monoculture



b) strip cropping

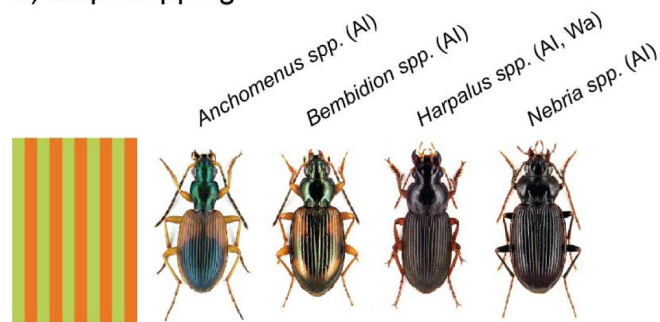


Fig. 2.

Ground beetle species associated with crop configuration (monoculture (a) versus strip cropping (b)).

Results obtained by generalized linear mixed models on the 12 most common genera of ground beetles with the four locations analysed separately (Table S5). Only those genera are given for which cropping system significantly influence activity density within at least one location ($\alpha = 0.05$). Locations where the genus had a higher activity density in one of the cropping systems are indicated between brackets (Al=Almere; Le = Lelystad; Va = Valthermond; Wa = Wageningen). For *Pterostichus*, data from Almere in 2020 was analysed separately, as models did not fit elsewhere. The asterisk "*" next to *Pterostichus* indicates that only in 2020 the difference between monoculture and strip cropping was significant. Photo credit: Ortwin Bleich, retrieved from: www.eurocarabidae.de .

(Karssemeijer et al., 2024 [↗](#)). Therefore, strip cropping may favour ground beetles that predate on, for instance, herbivore eggs, such as *Bembidion spp.* and *Anchomenus dorsalis*, which were more abundant in strip cropping (Finch, 1996 [↗](#)), and disadvantaging ground beetles that feed on larger prey, such as *Pterostichus melanarius*, which was less abundant in strip cropping. Multi-taxa evaluation of future strip cropping studies will be a valuable approach to increase understanding of biomass flows and trophic interactions within diverse agricultural systems.

The 15% increase in ground beetle species richness through strip cropping is mostly caused by an increase in species (relatively) common to agricultural fields, rather than an increase in rare species or species otherwise common in other habitat types. A keystone species here might be *Pterostichus melanarius*, the most common *Pterostichus* species in our samples. *Pterostichus melanarius* can be very dominant in pitfall traps, as we regularly found hundreds of individuals per trap. This relatively large ground beetle might compete with or predate on other ground beetle species (Roubinet et al., 2018 [↗](#)). *Pterostichus melanarius* is especially well-adapted to highly intensive agriculture (Turin, 2022 [↗](#), 2000 [↗](#)), which explains why it was more abundant in monocultures previously (Ditzler et al., 2021 [↗](#)). In turn, the reduced *P. melanarius* populations in strip cropping might allow other species to persist, such as *Harpalus spp.* and *Anchomenus dorsalis*. Therefore, strip cropping might facilitate ground beetle biodiversity by reducing the dominance of a common, competitive species.

Biodiversity gains in our study did in most cases not coincide with productivity losses (Table S6). Earlier studies on crop yield in the strip cropping fields in Lelystad and Wageningen during the years of pitfall trapping show a yield decrease when strip cropping cabbage (Carrillo-Reche et al., 2023 [↗](#)) and wheat (Ditzler et al., 2023 [↗](#)), whereas potato yield was unaffected by crop configuration (Ditzler et al., 2023 [↗](#)). In Almere bean and parsnip yields were higher in strip cropping, whereas oat and onion yield were lower (Juventia and van Apeldoorn, 2024 [↗](#)). In Valthermond crop productivity was similar for monocultures and strip cropping (Table S6). Therefore, biodiversity gains do not compromise agricultural production.

We show that changing crop configuration from monoculture to strip cropping, on average, enhances ground beetle richness by 15% and activity density by 30% within agricultural fields. These results show that strip cropping can lead to increases in biodiversity that approach those achieved by shifting from conventional to organic farming practices (+19% richness, Lichtenberg et al., 2017 [↗](#); +23% richness, Gong et al., 2022 [↗](#)) and by other in-field diversification measures, like hedgerows and flower strips (+23% richness, Lichtenberg et al., 2017 [↗](#); +24% richness, Beillouin et al., 2021 [↗](#)). While organic management or in-field diversification measures generally lead to lower productivity (Gong et al., 2022 [↗](#)), strip cropping can maintain crop productivity without taking land out of production (Campanelli et al., 2023 [↗](#); Juventia and van Apeldoorn, 2024 [↗](#); Van Oort et al., 2020 [↗](#)), although occasional yield reductions have been reported (Carrillo-Reche et al., 2023 [↗](#); Ditzler et al., 2023 [↗](#)). The biodiversity gain in our study was achieved without additional crop-level diversification strategies designed for biodiversity conservation, such as cover cropping or (flowering) companion plants. This increase in biodiversity stacks on top of already higher biodiversity achieved through organic management (Gong et al., 2022 [↗](#); Lichtenberg et al., 2017 [↗](#)). Furthermore, the estimated richness effects of strip cropping, i.e., no change at crop level and 15% increase at field level, are conservative because they only consider the pairwise comparison of biodiversity of two or three crops in a strip cropping configuration to monocultures, whereas the inclusion of more crops within strip cropping could further enhance ground beetle biodiversity. Also, the effect of strip cropping on ground beetle biodiversity might be more pronounced in conventional agriculture and at larger field sizes, as the relatively small-scale, organic fields that we studied might have already had a relatively high ground beetle biodiversity (Tschardt et al., 2021 [↗](#)). The ground beetle communities in our study were dominated by farmland and eurytopic species and contained only two rare species (Turin, 2022 [↗](#)). Future research could test whether the inclusion of other in-field diversification measures within

strip-cropped fields, such as the establishment of perennial semi-natural habitats (Risken et al., 2021 [↗](#); Sirami et al., 2019 [↗](#)) or uptake at larger spatial extents (Tscharncke et al., 2021 [↗](#)) would allow ground beetles with other habitat preferences to establish in agricultural fields.

Materials and methods

Study area

A multi-location study was conducted on four organic farms across the Netherlands (**Fig. 1a** [↗](#)). Three experimental farms were managed by Wageningen University & Research (Lelystad, Valthermond, Wageningen) and one commercial farm was managed by *Exploitatie Reservegronden Flevoland* (ERF B.V.) located in Almere. All four locations contained both strip cropping and monocultural crop fields, but differed in soil type, establishment year of the strip cropping experiment, number of crops grown, length of the crop rotation, number of sampled crops and sampling years, and farm and landscape characteristics such as percentage of on-farm semi-natural habitat (SNH), mean field size, and landscape configuration (**Fig. 1b** [↗](#), S10). The locations Almere and Lelystad were located in a homogeneous, open polder landscape characterized by intensive arable crop production and non-crop habitats consisting of grass margins, tree lines and watercourses. Valthermond was located in an open, reclaimed peat landscape with intensive arable crop production characterized by long and narrow fields separated by grassy margins and ditches and limited areas of woody elements. The site at Wageningen was located in a more complex landscape with smaller field sizes, and non-crop habitat consisting of woodlots, hedgerows, tree lines, ditches and farmyards.

Experimental lay-out

At Almere and Valthermond the crops in strip cropping were all grown alongside each other, whereas at Lelystad and Wageningen two alternating crops (crop pairs) were grown alongside each other (**Fig. 1b** [↗](#), S10). At each location, strip cropping and monoculture fields were always paired on the same experimental field (Fig. S10). At Almere eight different crops were grown in alternating strips of 6 m width, including celeriac (*Apium graveolens* var. *rapaceum*), broccoli (*Brassica oleracea* var. *italica*), oat (*Avena sativa*), onion (*Allium cepa*), parsnip (*Pastinaca sativa*), faba bean (*Vicia faba*), potato (*Solanum tuberosum*) and a mix of ryegrass and white clover referred to as grass-clover (*Lolium perenne*/*Trifolium repens*) (Fig. S10a). At Lelystad four different crop pairs were grown in alternating strips of 3 m width, including carrot (*Daucus carota* subsp. *sativus*) and onion, white cabbage (*Brassica oleracea* var. *capitata*) and wheat (*Triticum aestivum*), sugar beet (*Beta vulgaris*) and barley (*Hordeum vulgare*), and potato and ryegrass (Fig. S10b). At Valthermond eight different crops were grown in alternating strips of 6 m width, including potato, barley, barley mixed with broad bean (*Vicia faba*) in 2020, barley mixed with pea (*Pisum sativum*) in 2021, sweet corn (*Zea mays* convar. *saccharata* var. *rugosa*), sugar beet, common bean (*Phaseolus vulgaris*), and grass-clover (Fig. S10c). At Wageningen three different crop pairs were grown in alternating strips of 3 m width, including white cabbage and wheat (2019 – 2021) or oat (2022), barley and pumpkin (*Cucurbita maxima*, 2020 - 2022) or bare soil due to crop failure (2019), and potato and ryegrass (Fig. S10d). The crop combinations and neighbors were selected based on literature, expert knowledge and experience of functionality in terms of expected advantages for yield and pest and disease control. Large scale monoculture plots (0.25 ha to 2.30 ha) served as reference, hereafter referred to as monoculture (Fig. S10). At Lelystad, Valthermond and Wageningen not each crop grown in strips was present as monoculture in each year, but only those crops for which a monoculture was present were sampled (**Fig. 1k** [↗](#), Table S7). All fields were managed according to organic regulations, yet at each location fertilization and weed management reflected regional practices and were adjusted to local soil conditions. Flower strips were sown within the experimental fields at Almere and Valthermond (Fig. S10, Table S8).

Sampling

The ground beetle community was sampled using pitfall traps in all crops for which both a monoculture and strip cropping field were present, at each location and in multiple rounds per year between March and September. The sampled crops, number of rounds and number of pitfalls differed per year and per location (Table S7). Pitfall traps consisted of a transparent plastic cup (9.2 cm diameter, 14 cm height) placed in the soil so the top of the cup was level with the soil surface. We did not use funnels inside the cups. Pitfalls were filled with approximately 100 ml water mixed with non-perfumed soap (around 3 cm high water level) and covered with a black plastic rain guard (12.5 cm diameter) around 2 to 5 cm above the soil surface. Pitfall traps were placed in the center of a strip (1.5 – 3 meters from the edges of the strip), and at least 10 meters from field edges at fixed locations at different moments in a year. Furthermore, traps were mostly placed at equal distances from the field edges in strip-cropped and monocultural fields. At Almere and Valthermond, the number of pitfalls in monocultures and strip cropping was the same. At Wageningen and Lelystad, the number of pitfalls in strip cropped fields was usually lower than in the monoculture because the area of the strips was only half of that of the monocultures.

Ground beetle identification

Ground beetles were identified up to species level for Almere in 2021 and 2022, Lelystad and Wageningen. At Almere in 2020, identification was kept at the genus level due to an extremely high abundance of *Pterostichus melanarius/niger* and *Poecilus cupreus/versicolor*, and the associated high time investment for identification. At Valthermond, due to labour constraints, we chose to consider species that are complicated to distinguish only up to genus level (like *Poecilus cupreus/versicolor*, *Harpalus* spp. other than *H. rufipes* and *H. affinis*, *Bembidion* spp.). For all analyses, year series were made, in which all ground beetle catches from the same pitfall trap were pooled per year.

Statistical analyses

We used R, version 4.2.2 for all statistical analyses.

Effect of crop configuration on field-level richness and activity density

To analyse the difference in species richness and activity density between monocultures and strip cropping configurations at field level, we used rarefaction of samples within the same field. To rarify to an equal sampling intensity, we calculated the average cumulative number of species or individuals within x year series, where x is the largest number of year series available for the crop configuration comparison (correcting for unequal sampling between monocultures and strip cropping, or for missing samples). Next, we calculated the relative change due to strip cropping by subtracting the number of species or individuals found in the monoculture field from the number found in the strip cropping field and then dividing the result by the number of species or individuals in the monoculture. This gave the relative change centered around zero, where negative values indicated higher richness in monocultures and positive values higher richness in strip cropping. We then analysed this data using Generalized Linear Mixed Models (GLMM) with a Gaussian distribution and assessed whether the intercept deviated significantly from zero. As random variables, we used location and year, with year nested in location. We ran these analyses using a dataset that included all comparisons among monocultural fields and strip cropping fields of all locations. However, as in this case the strip cropped field was compared with both monocultural fields of the corresponding crop pair, we also tested the effect of field-level richness using only one monoculture per strip cropped field. To obtain a conservative estimate of the effect of strip cropping, we chose the monocultural fields with the highest taxonomic richness or activity density among the constitutive crops of the strip cropping fields. Generalized linear models were run using the `glmmTMB` package (Brooks et al., 2017 [↗](#)) and tested for model fit using the `DHARMA` package (Hartig, 2018 [↗](#)).

Effect of crop configuration on crop-level biodiversity

To quantify biodiversity we used five variables: (1) activity density, the total number of ground beetles found per year series; (2) taxonomic richness, the total number of species or genera (lowest taxonomic level available) found per year series; (3) the inverse Simpson index, the inverse of the sum of proportions of different species over the total abundance (Simpson, 1949); (4) absolute evenness, the number of effective species calculated by dividing the inverse Simpson index by the taxonomic richness (Williams, 1964); and (5) Shannon entropy (Shannon and Weaver, 1949). We chose absolute evenness as our measure for evenness, as this method removes the richness component from the inverse Simpson index and adheres to all requirements for an evenness index (Smith and Wilson, 1996; Tuomisto, 2012). We included both the inverse Simpson index and Shannon entropy as the former is more sensitive to changes in evenness and the latter to species richness (DeJong, 1975).

To analyse the effect of crop configuration on total ground beetle activity density, taxonomic richness, evenness, inverse Simpson index and Shannon entropy we used GLMM. We constructed models for each response variable, using data from all four locations. In these models we included crop configuration (monoculture or strip cropping) as a fixed factor. We included location, year and crop as nested random variables in these models, with crop nested in year and year in location. To quantify and visualize the variation in responses between locations, years and crops, we ran generalized linear models (GLM) with a variable that combined these three variables into one, which was also included as a fixed factor. Here, we also included the interaction between crop configuration and the combined variable for crop, location and year. For the model on activity density we used the negative binomial distribution (log link function), as this was count data; for richness, evenness and Shannon entropy we used Gaussian distribution; and for the inverse Simpson index we used a gamma distribution (inverse link function) as a Gaussian distribution did not fit well. We used the ‘DHARMA’ package to validate model assumptions. We used estimated marginal means to assess differences between monoculture and strip cropping.

To analyse the effect of crop configuration on the activity density of the twelve most common ground beetle genera, we used another GLMM with negative binomial distribution. Fixed effects were crop configuration, location and their interactions, and crop nested in year was added as a random effect. As certain genera only occurred at specific locations, we only included locations where the genus was found. All models were tested for model fit using the DHARMA package. We used estimated marginal means to assess differences between monoculture and strip cropping.

Community composition of crop configurations and crops

To assess whether crop configurations have distinct ground beetle communities we used permanova with a Hellinger transformation (with 999 permutations) using the “vegan” package (Oksanen et al., 2013 [DOI](#)). We used a Hellinger transformation to give more weight to rarer species, thus accounting for highly abundant species in some samples (such as *P. melanarius* and *H. rufipes*). We only included species that occurred in at least 3% of the pitfall samples to avoid strong influence of very rare species. We used data from locations and years where pitfall catches had been identified to species level (Almere 2021/22, Lelystad and Wageningen). We considered four models, one for all locations combined and one for each location separately. In all models we included crop configuration and the nested variables of location (whenever applicable), year and crop as fixed factors. We also analysed the interaction between crop configuration and this nested structure of location, year and crop. To visualize these results, we used non-metric multidimensional scaling (NMDS) on Hellinger transformed data. To show whether crop configurations have distinct ground beetle communities, we used redundancy analysis (RDA) on Hellinger transformed data. Here, we again conducted four analyses, one for all locations combined and one for each of the three considered locations. We only used crop configuration as a

predictor to force RDA to show any change in ground beetle community associated with crop configuration. As such, only one RDA axis was created per model, which was plotted against the first principal component describing the residual variation.

Due to the large influence of location and year on ground beetle communities, visualizing any general effects of strip cropping on these communities using all available data was challenging. To address this, we conducted RDA and visualized the effect of crop configuration on a subset of the data from one location and one year. We chose the data from Wageningen in 2021 and 2022 for this analysis because it provided a set-up where strip cropping of two crops could be compared with their constituent monocultures within the same experimental fields. Here, we used both crop configuration, crop and their interaction term as explanatory variables for each crop pair separately. To analyse whether ground beetle communities significantly differed among combinations of crops and crop configurations we ran pairwise permanova on all three fields and two years separately using the “pairwiseAdonis” package (Martinez Arbizu, 2020 [DOI](#)).

Availability of data and materials

Data and scripts are publicly available from 4TU.ResearchData via the following DOI: <https://doi.org/10.4121/bcf78320-aaa6-428f-acf6-2eb436baa13e> (Croijmans et al., 2025 [DOI](#)).

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Ethics approval

This research was not reviewed by an institutional or governmental regulatory body as the work was performed on invertebrates.

Additional files

Supplemental tables and figures [🔗](#)

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University of Neuchâtel, Neuchâtel, Switzerland

Reviewer #3 (Public review):

Summary: In this paper the authors examined the effects of strip cropping, a relatively new agricultural technique of alternating crops in small strips of several meters wide, on ground beetle diversity. The results show an increase in species diversity (i.e. abundance and species richness) of the ground beetle communities compared to monocultures.

Strengths: The article is well written; it has an easily readable tone of voice without too much jargon or overly complicated sentence structure. Moreover, as far as reviewing the models in depth without raw data and R scripts allows, the statistical work done by the authors looks good. They have well thought out how to handle heterogeneous, unbalanced and taxonomically unspecific yet spatially and temporarily correlated field data. The models applied and the model checks performed are appropriate for the data at hand. Combining RDA and PCA axes together is a nice touch. Moreover, after the first round of reviews, the authors have done a great job at rewriting the paper to make it less overstated, more relevant to the data at hand and more solid in the findings. Many of the weaknesses noted in the first review have been dealt with. The overall structure of the paper is good, with a clear introduction, hypotheses, results section and discussion.

Weaknesses: The weaknesses that remain are mainly due to a difficult dataset and choices that could have stressed certain aspects more, like the relationship between strip cropping and intercropping. The mechanistic understanding of strip cropping is what is at stake here. Does strip cropping behave similar to intercropping, a technique which has been proven to be beneficial to biodiversity because of added effects due to increased resource efficiency and greater plant species richness.

Unfortunately, the authors do not go into this in the introduction or otherwise and simply state that they consider strip cropping a form of intercropping.

I also do not like the exclusive focus on percentages, as these are dimensionless. I think more could have been done to show underlying structure in the data, even after rarefaction.

A further weakness is a limited embedding into the larger scientific discourses other than providing references. But this may be a matter of style and/or taste

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Author response:

The following is the authors' response to the original reviews.

We thank all reviewers for the highly detailed review and the time and effort which has been invested in this review. It is clear from the reviews that we've had the privilege to have our work extensively and thoroughly checked by knowledgeable experts, for which we are very grateful. We have read their perspectives, questions and suggested improvements with great interest. We have reflected on the public review in detail and have included detailed responses below. First, we would like to respond to four main issues pointed out by the editor and reviewers:

(1) Lack of yield data in the manuscript: Yield data has been collected in most of the sites and years of our study, and these have already been published and cited in our manuscript. In the appendix of our manuscript, we included a table with yield data for the sites and years in

which the beetle diversity was studied. These data show that strip cropping does not cause a systematic yield reduction.

(2) Sampling design clarification: Our paper combines data from trials conducted at different locations and years. On the one hand this allows an analysis of a comprehensive dataset, but on the other hand in some cases this resulted in variations in how data were collected or processed (e.g. taxonomic level of species identification). We have added more details to the sections on sampling design and data analysis to increase clarity and transparency.

(3) Additional data analysis: In the revised manuscript we present an analysis on the responses of abundances of the 12 most common ground beetle genera to strip cropping. This gives better insight in the variation of responses among ground beetle taxa.

(4) Restrict findings to our system: We nuanced our findings further and focused more on the implications of our data on ground beetle communities, rather than on agrobiodiversity in a broader sense.

Below we also respond to the editor and reviewers in more detail.

Reviewing Editor Comments:

(1) You only have analyzed ground beetle diversity, it would be important to add data on crop yields, which certainly must be available (note that in normal intercropping these would likely be enhanced as well).

Most yield data have been published in three previous papers, which we already cited or cite now (one was not yet published at the time of submission). Our argumentation is based on these studies. We had also already included a table in the appendix that showed the yield data that relates specifically to our locations and years of measurement. The finding that strip cropping does not majorly affect yield is based on these findings. We revised the title of our manuscript to remove the explicit focus on yield.

(2) Considering the heterogeneous data involving different experiments it is particularly important to describe the sampling design in detail and explain how various hierarchical levels were accounted for in the analysis.

We agree that some important details to our analysis were not described in sufficient detail. Especially reviewer 2 pointed out several relevant points that we did account for in our analyses, but which were not clear from the text in the methods section. We are convinced that our data analyses are robust and that our conclusions are supported by the data. We revised the methods section to make our approach clearer and more transparent.

(3) In addition to relative changes in richness and density of ground beetles you should also present the data from which these have been derived. Furthermore, you could also analyze and interpret the response of the different individual taxa to strip cropping.

With our heterogeneous dataset it was quite complicated to show overall patterns of absolute changes in ground beetle abundance and richness, especially for the field-level analyses. As the sampling design was not always the same and occasionally samples were missing, the number of year series that made up a datapoint were different among locations and years. However, we always made sure that for the comparison of a paired monoculture and strip cropping field, the number of year series was always made equal through rarefaction. That is, the number of ground beetle(s) (species) are always expressed as the number per 2 to 6 samples. Therefore, we prefer to stick to relative changes as we are convinced that this gives a fairer representation of our complex dataset.

We agree with the second point that both the editor and several reviewers pointed out. The indicator species analyses that we used were biased by rare species, and we now omit this analysis. Instead, we included a GLM analysis on the responses of abundances of the 12 most common ground beetle genera to strip cropping. We chose for genera here (and not species) as we could then include all locations and years within the analyses, and in most cases a genus was dominated by a single species (but notable exceptions were *Amara* and *Harpalus*, which were often made up of several species). We illustrate these analyses still in a similar fashion as we did for the indicator species analysis.

(4) Keep to your findings and don't overstate them but try to better connect them to basic ecological hypotheses potentially explaining them.

After careful consideration of the important points that reviewers point out, we decided to nuance our reasoning about biodiversity conservation along two key lines: (1) the extent to which ground beetles can be indicators of wider biodiversity changes; and (2) our findings that are not as straightforward positive as our narrative suggests. We still believe that strip cropping contributes positively to carabid communities, and have carefully checked the text to avoid overstatements.

Reviewer #1 (Public review):

Summary:

This study demonstrates that strip cropping enhances the taxonomic diversity of ground beetles across organically-managed crop systems in the Netherlands. In particular, strip cropping supported 15% more ground beetle species and 30% more individuals compared to monocultures.

Strengths:

A well-written study with well-analyzed data of a complex design. The data could have been analyzed differently e.g. by not pooling samples, but there are pros and cons for each type of analysis and I am convinced this will not affect the main findings. A strong point is that data were collected for 4 years. This is especially strong as most data on biodiversity in cropping systems are only collected for one or two seasons. Another strong point is that several crops were included.

We thank reviewer 1 for their kind words and agree with this strength of the paper. The paper combines data from trials conducted at different locations and years. On the one hand this allows an analysis of a comprehensive dataset, but on the other hand in some cases there were slight variations in how data were collected or processed (e.g. taxonomic level of species identification).

Weaknesses:

This study focused on the biodiversity of ground beetles and did not examine crop productivity. Therefore, I disagree with the claim that this study demonstrates biodiversity enhancement without compromising yield. The authors should present results on yield or, at the very least, provide a stronger justification for this statement.

We acknowledge that we indeed did not formally analyze yield in our study, but we have good reason for this. The claim that strip cropping does not compromise yield comes from several extensive studies (Juventia & van Apeldoorn, 2024; Ditzler et al., 2023; Carillo-Reche et al., 2023) that were conducted in nearly all the sites and years that we included in our study. We chose not to include formal analyses of productivity for two key reasons: (1) a yield

analysis would duplicate already published analyses, and (2) we prefer to focus more on the ecology of ground beetles and the effect of strip cropping on biodiversity, rather than diverging our focus also towards crop productivity. Nevertheless, we have shown the results on yield in Table S6 and refer extensively to the studies that have previously analyzed this data (line 203-207, 217-221).

Reviewer #1 (Recommendations for the authors):

This is a well-written study on the effects of strip cropping on ground-beetle diversity. As stated above the study is well analyzed, presented, and written but you should not pretend that you analyzed yield e.g. lines 25-27 "We show that strip cropping...enhance ground beetle biodiversity without incurring major yield loss."

We understand the confusion caused by this sentence, and it was never our intention to give the impression that we analyzed yield losses. These findings were based on previous research by ourselves and colleagues, and we have now changed the sentence to reflect this (line 25-27).

I think you assume that yield does not differ between strip cropping and monoculture. I am not sure this is correct as one crop might attract pests or predators spilling over to the other crop. I am also not sure if the sowing and harvest of the crop will come with the same costs. So if you assume this, you should only do it in the main manuscript and not the abstract, to justify this better.

With three peer-reviewed papers on the same fields as we studied, we can convincingly state that strip cropping in organic agriculture generally does not result in major yield loss, although exceptions exist, which we refer to in the discussion.

In the introduction lines 28-43, you refer to insect biomass decline. I wonder if you would like to add the study of Loboda et al. 2017 in Ecography. It seems not fitting as it is from the Arctic but also the other studies you cite are not only coming from agricultural landscapes and this study is from the same time as the Hallmann et al. 2017 study and shows a decline in flies of 80%

We have removed the sentence that this comment refers to, to streamline the introduction more.

Lines 50-51. You only have one citation for biodiversity strategies in agricultural systems. I suggest citing Mupepele et al. 2021 in TREE. This study refers to management but also the policies and societal pressures behind it.

We have added this citation and a recent paper by Cozim-Melges et al. (2024) here (line 49-52).

In the methods, I am missing a section on species identifications. This would help to understand why you used "taxonomic richness".

Thanks for pointing this out. We have now included a new section on ground beetle identification (line 304-309 in methods).

Figure 1 is great and I like that you separated the field and crop-level data, although there is no statistical power for the crop-specific data. I personally would move k to the supplements. It is very detailed and small and therefore hard to read

We chose to keep figure 1k, as in our view it gives a good impression of the scale of the experiment, the number of crops included and the absolute numbers of caught species.

Reviewer #2 (Public review):

Summary:

The authors aimed to investigate the effects of organic strip cropping on carabid richness and density as well as on crop yields. They find on average higher carabid richness and density in strip cropping and organic farming, but not in all cases.

We did not intend to investigate the effect of strip cropping on crop yields, but rather place our work in the framework of earlier studies that already studied yield. All the monocultures and strip cropping fields were organic farms. Our findings thus compare crop diversity effects within the context of organic farming.

Strengths:

Based on highly resolved species-level carabid data, the authors present estimates for many different crop types, some of them rarely studied, at the same time. The authors did a great job investigating different aspects of the assemblages (although some questions remain concerning the analyses) and they present their results in a visually pleasing and intuitive way.

We appreciate the kind words of reviewer 2 and their acknowledgement of the extensiveness of our dataset. In our opinion, the inclusion of many different crops is indeed a strength, rarely seen in similar studies; and we are happy that the figures are appreciated.

Weaknesses:

The authors used data from four different strip cropping experiments and there is no real replication in space as all of these differed in many aspects (different crops, different areas between years, different combinations, design of the strip cropping (orientation and width), sampling effort and sample sizes of beetles (differing more than 35 fold between sites; L 100f); for more differences see L 237ff). The reader gets the impression that the authors stitched data from various places together that were not made to fit together. This may not be a problem per se but it surely limits the strength of the data as results for various crops may only be based on small samples from one or two sites (it is generally unclear how many samples were used for each crop/crop combination).

The paper indeed combines data from trials conducted at different locations and years. On the one hand this allows an analysis of a comprehensive dataset, but on the other hand in some cases there were slight differences in the experimental design. At the time that we did our research, there were only a handful of farmers that were employing strip cropping within the Netherlands, which greatly reduced the number of fields for our study. Therefore, we worked in the sites that were available and studied as many crops on these sites. Since there was variation in the crops grown in the sites, for some crops we have limited replication. In the revision we have explained this more clearly (line 297-300).

One of my major concerns is that it is completely unclear where carabids were collected. As some strips were 3m wide, some others were 6m and the monoculture plots large, it can be expected that carabids were collected at different distances from the plot edge. This alone, however, was conclusively shown to affect carabid assemblages dramatically and could easily outweigh the differences shown here if not accounted for in the models

(see e.g. Boetzi et al. (2024) or Knapp et al. (2019) among many other studies on within field-distributions of carabids).

Point well taken. Samples were always taken at least 10 meters into the field, and always in the middle of the strip. This would indeed mean that there is a small difference between the 3- and 6m wide strips regarding distance from another strip, but this was then only a difference of 1.5 to 3 meters from the edge. A difference that, based on our own extensive experience with ground beetle communities, will not have a large impact on the findings of ground beetles. The distance from field/plot edges was similar between monocultures and strip cropped fields. We present a more detailed description of the sampling design in the methods of the revised manuscript (line 294-297).

The authors hint at a related but somewhat different problem in L 137ff - carabid assemblages sampled in strips were sampled in closer proximity to each other than assemblages in monoculture fields which is very likely a problem. The authors did not check whether their results are spatially autocorrelated and this shortcoming is hard to account for as it would have required a much bigger, spatially replicated design in which distances are maintained from the beginning. This limitation needs to be stated more clearly in the manuscript.

To be clear, this limitation relates to the comparison that we did for the community compositions of ground beetles in two crops either in strip cropping or monocultures. In this case, it was impossible to avoid potential autocorrelation due to our field design. We also acknowledge this limitation in the results section (line 130-133). However, for our other analyses we corrected for spatial autocorrelation by including variables per location, year and crop. This grouped samples that were spatially autocorrelated. Therefore, we don't see this as a discrepancy of our other analyses.

Similarly, we know that carabid richness and density depend strongly on crop type (see e.g. Toivonen et al. (2022)) which could have biased results if the design is not balanced (this information is missing but it seems to be the case, see e.g. Celeriac in Almere in 2022).

We agree and acknowledge that crop type can influence carabid richness and density, which is why we have included variables to account for differences caused by crops. However, we did not observe consistent differences between crops in how strip cropping affected ground beetle richness and density. Therefore, we don't think that crop types would have influenced our conclusions on the overall effect of strip cropping.

A more basic problem is that the reader neither learns where traps were located, how missing traps were treated for analyses how many samples there were per crop or crop combination (in a simple way, not through Table S7 - there has to have been a logic in each of these field trials) or why there are differences in the number of samples from the same location and year (see Table S7). This information needs to be added to the methods section.

Point well taken. We have clarified this further in the revised manuscript (line 294-301, 318-322). As we combined data from several experimental designs that originally had slightly different research questions, this in part caused differences between numbers of rounds or samples per crop, location or year.

As carabid assemblages undergo rapid phenological changes across the year, assemblages that are collected at different phenological points within and across years cannot easily be compared. The authors would need to standardize for this and make

sure that the assemblages they analyze are comparable prior to analyses. Otherwise, I see the possibility that the reported differences might simply be biased by phenology.

We agree and we dealt with this issue by using year series instead of using individual samples of different rounds. This approach allowed us to get a good impression of the entire ground beetle community across seasons. For our analyses we had the choice to only include data from sampling rounds that were conducted at the same time, or to include all available data. We chose to analyze all data, and made sure that the number of samples between strip cropping and monoculture fields per location, year and crop was always the same by pooling and rarefaction.

Surrounding landscape structure is known to affect carabid richness and density and could thus also bias observed differences between treatments at the same locations (lower overall richness => lower differences between treatments). Landscape structure has not been taken into account in any way.

We did not include landscape structure as there are only 4 sites, which does not allow a meaningful analysis of potential effects landscape structure. Studying how landscape interacts with strip cropping to influence insect biodiversity would require at least, say 15 to 20 sites, which was not feasible for this study. However, such an analysis may be possible in an ongoing project (CropMix) which includes many farms that work with strip cropping.

In the statistical analyses, it is unclear whether the authors used estimated marginal means (as they should) - this needs to be clarified.

In the revised manuscript we further clarified this point (line 365-366, 373-374).

In addition, and as mentioned by Dr. Rasmann in the previous round (comment 1), the manuscript, in its current form, still suffers from simplified generalizations that 'oversell' the impact of the study and should be avoided. The authors restricted their analyses to ground beetles and based their conclusions on a design with many 'heterogeneities' - they should not draw conclusions for farmland biodiversity but stick to their system and report what they found. Although I understand the authors have previously stated that this is 'not practically feasible', the reason for this comment is simply to say that the authors should not oversell their findings.

In the revised manuscript, we nuanced our findings by explaining that strip cropping is a potentially useful tool to support ground beetle biodiversity in agricultural fields (line 33-35).

Reviewer #2 (Recommendations for the authors):

In addition to the points stated under 'Weaknesses' above, I provide smaller comments and recommendations:

Overall comments:

(i) The carabid images used in the figures were created by Ortwin Bleich and are copyrighted. I could not find him accredited in the acknowledgements; the figure legends simply state that the images were taken from his webpage. Was his permission obtained? This should be stated.

We have received written permission from Ortwin Bleich for using his pictures in our figures, and have accredited him for this in the acknowledgements (line 455-456).

(ii) There is a great confusion in the field concerning terminology. The authors here use intercropping and strip cropping, a specific form of intercropping, interchangeably. I advise the authors to stick to strip cropping as it is more precise and avoids confusion with other forms of intercropping.

We agree with the definitions given by reviewer 2 and had already used them as such in the text. We defined strip cropping in the first paragraph of the introduction and do not use the term “intercropping” after this definition to avoid confusion.

Comments to specific lines:

Line 19: While this is likely true, there is so far not enough compelling evidence for such a strong statement blaming agriculture. Please rephrase.

Changed the sentence to indicate more clearly that it is one of the major drivers, but that the “blame” is not solely on agriculture (line 18-19).

Line 22: Is this the case? I am aware of strip cropping being used in other countries, many of them in Europe. Why the focus on 'Dutch'?

Indeed, strip cropping is now being pioneered by farmers throughout Europe. However in the Netherlands, some farmers have been pioneering strip cropping already since 2014. We have added this information to indicate that our setting is in the Netherlands, and as in our opinion it gives a bit more context to our manuscript.

Line 24: I would argue that carabids are actually not good indicators for overall biodiversity in crop fields as they respond in a very specific way, contrasting with other taxa. It is commonly observed that carabids prefer more disturbed habitats and richness often increases with management intensity and in more agriculturally dominated landscapes - in stark contrast to other taxa like wild bees or butterflies.

We have reworded this sentence to reflect that they are not necessarily indicators of wide agricultural biodiversity, but that they do hold keystone positions within food webs in agricultural systems (line 23-25).

Line 31: This statement here is also too strong - carabids are not overall biodiversity and patterns found for carabids likely differ strongly from patterns that would be observed in other taxa. This study is on carabids and the conclusion should thus also refer to these in order to avoid such over-simplified generalizations.

We agree and have nuanced this sentence to indicate that our findings are only on ground beetles (line 33-35). However, we would like to point out that the statement that “patterns found for carabids likely differ strongly from patterns that would be observed in other taxa” assumes a disassociation between carabids and other taxa.

Line 41: I am sure the authors are aware of the various methodological shortcomings of the dataset used in Hallmann et al. (2017) which likely led to an overestimation of the actual decline. Analysing the same data, Müller et al. (2023) found that weather can explain fluctuations in biomass just as well as time. I thus advise not putting too much focus on these results here as they seem questionable.

We have removed this sentence to streamline the introduction, thus no longer mentioning the percentages given by Hallmann et al. (2017).

Line 46: Surely likely but to my knowledge this is actually remarkably hard to prove. Instead of using the IPBES report here that simply states this as a fact, it would be better to see some actual evidence referenced.

We removed IPBES as a source and changed this for Dirzo et al. (2014), a review that shows the consequences of biodiversity decline on a range of different ecosystem services and ecological functions (line 45-47).

Line 52ff: I am not sure whether this old land-sparing vs. land-sharing debate is necessary here. The authors could simply skip it and directly refer to the need of agricultural areas, the dominating land-use in many regions, to become more biodiversity-friendly. It can be linked directly to Line 61 in my opinion which would result in a more concise and arguably stronger introduction.

After reconsidering, we agree with reviewer 2 that this section was redundant and we have removed the lines on land-sparing vs land-sharing.

Line 59: Just a note here: this argument is not meaningful when talking about strip cropping in the Netherlands as there is virtually no land left that could be converted (if anything, agricultural land is lost to construction). The debate on land-use change towards agriculture is nowadays mostly focused on the tropics and the Global South.

We argue that strip cropping could play an important role as a measure that does not necessarily follow the trade-off between biodiversity and agriculture for a context beyond the Netherlands (line 52-58).

Line 69: Does this statement really need 8 references?

Line 71: ... and this one 5 additional ones?

We have removed excess references in these two lines (line 62-66).

Line 74: But also likely provides the necessary crop continuity for many crop pests - the authors should keep in mind that when practitioners read agricultural biodiversity, they predominantly think of weeds and insect pests.

We agree with reviewer 2 that agricultural biodiversity is still a controversial topic. However, as the focus in this manuscript is more on biodiversity conservation, rather than pest management, we prefer to keep this sentence as is. In other published papers and future work we focus more on the role of strip cropping for pest management.

Line 83: Consider replacing 'moments' maybe - phenological stages or development stages?

Although we understand the point of reviewer 2, we prefer to keep it at moments, as we did not focus on phenological stages and we only wanted to say that we set pitfall traps at several moments throughout the year. However, by placing the pitfall traps at several moments throughout the year, we did capture several phenological stages.

Line 86: Not only farming practices - there are also massive fluctuations between years in the same crop with the same management due to effects of the weather in the previous reproductive season. Interpreting carabid assemblage changes is therefore not straightforward.

We absolutely agree that interpreting carabid assemblage is not straightforward, but as we did not study year or crop legacy effects we chose to keep this sentence to maintain focus on our research goals.

| Line 88: 'ecolocal'?

Typo, should have been ecological. Changed (line 81).

| Line 90: 'As such, they are often used as indicator group for wider insect diversity in agroecosystems' - this is the third repetition of this statement and the second one in this paragraph - please remove. Having worked on carabids extensively myself, I also think that this is not the true reason - they are simply easy to collect passively.

We agree with the reviewer and have removed this sentence.

| Line 141: I have doubts about the value of the ISA looking at the results. *Anchomenus dorsalis* is a species extremely common in cereal monoculture fields in large parts of Europe, especially in warmer and drier conditions (*H. griseus* was likely only returned as it is generally rare and likely only occurred in few plots that, by chance, were strip-cropped). It can hardly be considered an indicator for diverse cropping systems but it was returned as one here (which I do not doubt). This often happens with ISA in my experience as they are very sensitive to the specific context of the data they are run on. The returned species are, however, often not really useable as indicators in other contexts. I thus believe they actually have very limited value. Apart from this, we see here that both monocultures and strip cropping have their indicators, as would likely all crop types. I wonder what message we would draw from this ...

On close reconsideration, we agree with the reviewer that the ISAs might have been too sensitive to rare species that by chance occur in one of two crop configurations. To still get an idea on what happens with specific ground beetle groups, we chose to replace the ISAs with analyses on the 12 most common ground beetle genera. For this purpose we have added new sections to the methods (line 368-374) and results (line 135-143), replaced figure 2 and table S5, and updated the discussion (line 182-200).

| Line 165: Carabid activity is high when carabids are more active. Carabids can be more active either when (i) there are simply more carabid individuals or /and (ii) when they are starved and need to search more for prey. More carabid activity does thus not necessarily indicate more individuals, it can indicate that there is less prey. This aspect is missing here and should be discussed. It is also not true that crop diversification always increases prey biomass - especially strip cropping has previously been shown to decrease pest densities (Alarcón-Segura et al., 2022). Of course, this is a chicken-egg problem (less pests => less carabids or more carabids => less pests ?) ... this should at least be discussed.

We have rewritten this paragraph to further discuss activity density in relation to food availability (line 175-185).

| Line 178: These species are not exclusively granivorous - this speculation may be too strong here.

| Line 185: true for all but *C. melanocephalus* - this species is usually more associated with hedgerows, forests etc.

After removing the ISA's, we also chose to remove this paragraph and replace it with a paragraph that is linked to the analyses on the 12 most common genera (line 182-200).

Line 202: These statements are too strong for my taste - the authors should add an 'on average' here. The data show that they likely do not always enhance richness by 15 % and as the authors state, some monocultures still had higher richness and densities.

“on average” added (line 211)

Line 203: 'can lead' - the authors cannot tell based on their results if this is always true for all taxa.

Changed to “can lead” (line 213)

Line 205: What is 'diversification' here?

This concerns measures like hedgerows or flower strips. We altered the sentence to make this clearer (line 215-216).

Line 208: Does this statement need 5 references? (as in the introduction, the reader gets the impression the authors aimed to increase the citation count of other articles here).

We have removed excess references (line 219-221).

Line 222: How many are 'a few'? Maybe state a proportion.

We only found two species, we've changed the sentence accordingly (line 232-233).

Line 224: As stated above, I would not overstress the results of the ISAs - the authors stated themselves that the result for A. dorsalis is likely only based on one site ...

We removed this sentence after removing the ISAs.

Line 305: I think there is an additional nested random level missing - the transect or individual plot the traps were located in (or was there only one replicate for each crop/strip in each experiment)? Hard to tell as the authors provide no information on the actual sample sizes.

Indeed, there was one field or plot per cropping system per crop per location per year from which all the samples were taken. Therefore the analysis does not miss a nested random level. We provided information on sample sizes in Table S7.

Line 314ff: The authors describe that they basically followed a (slightly extended) Chao-Hill approach (species richness, Shannon entropy & inverse Simpson) without the sampling effort / sample completeness standardization implemented in this approach and as a reader I wonder why they did not simply just use the customary Chao-Hill approach.

We were not aware of the Chao-Hill approach, and we see it as a compliment that we independently came up with an approach similar to a now accepted approach.

Line 329: Unclear what was nested in what here - location / year / crop or year / location / crop ?

For the crop-level analyses, the nested structure was location > year > crop. This nested structure was chosen as every location was sampled across different years and (for some locations) the crops differed among years. However, as we pooled the samples from the same field in the field-level analyses, using the same random structure would have resulted in each individual sampling unit being distinguished as a group. Therefore, the random structure here was only location > year. We explain this now more clearly in lines 329 and 355-357.

Line 334: I can see why the authors used these distributions but it is presented here without any justification. As a side note: Gamma (with log link) would likely be better for the Shannon model as well (I guess it cannot be 0 or negative ...).

We explain this now better in lines 360-364.

Line 341: Why Hellinger and not simply proportions?

We used Hellinger transformation to give more weight to rarer species. Our pitfall traps were often dominated by large numbers of a few very abundant / active species. If we had used proportions, these species would have dominated the community analyses. We clarified this in the text (line 379-381).

Line 348: An RDA is constrained by the assumptions / model the authors proposed and "forces" the data into a spatial ordination that resembles this model best. As the authors previously used an unconstrained PERMANOVA, it would be better to also use an NMDS that goes along with the PERMANOVA.

The initial goal of the RDA was not to directly visualize the results of the PERMANOVA, but to show whether an overall crop configuration effect occurred, both for the whole dataset and per location. We have now added NMDS figures to link them to the PERMANOVA and added these to the supplementary figures (fig S6-S8). We also mention this approach in the methods section (line 387-390).

Line 355f: This is also a clear indication of the strong annual fluctuations in carabid assemblages as mentioned above.

Indeed.

Line 361: 'pairwise'.

Typo, we changed this.

Line 362: reference missing.

Reference added (line 405)

References

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Reviewer #3 (Public review):

Summary:

In this paper, the authors made a sincere effort to show the effects of strip cropping, a technique of alternating crops in small strips of several meters wide, on ground beetle diversity. They state that strip cropping can be a useful tool for bending the curve of biodiversity loss in agricultural systems as strip cropping shows a relative increase in species diversity (i.e. abundance and species richness) of the ground beetle communities compared to monocultures. Moreover, strip cropping has the added advantage of not having to compromise on agricultural yields.

Strengths:

The article is well written; it has an easily readable tone of voice without too much jargon or overly complicated sentence structure. Moreover, as far as reviewing the models in depth without raw data and R scripts allows, the statistical work done by the authors looks good. They have well thought out how to handle heterogenous, yet spatially and temporarily correlated field data. The models applied and the model checks performed are appropriate for the data at hand. Combining RDA and PCA axes together is a nice touch.

We thank reviewer 3 for their kind words and appreciation for the simple language and analysis that we used.

Weaknesses:

The evidence for strip cropping bringing added value for biodiversity is mixed at best. Yes, there is an increase in relative abundance and species richness at the field level, but it is not convincingly shown this difference is robust or can be linked to clear structural and hypothesised advantages of the strip cropping system. The same results could have been used to conclude that there are only very limited signs of real added value of strip cropping compared to monocultures.

Point well taken. We agree that the effect of strip cropping on carabid beetle communities are subtle and we nuanced the text in the revised version to reflect this. See below for more

details on how we revised the manuscript to reflect this point.

There are a number of reasons for this:

(1) Significant differences disappear at crop level, as the authors themselves clearly acknowledge, meaning that there are no differences between pairs of similar crops in the strip cropping fields and their respective monoculture. This would mean the strips effectively function as "mini-monocultures".

This is indeed in line with our conclusions. Based on our data and results, the advantages of strip cropping seem mostly to occur because crops with different communities are now on the same field, rather than that within the strips you get mixtures of communities related to different crops. We discussed this in the first paragraph of the discussion in the original submission (line 161-164).

The significant relative differences at the field level could be an artifact of aggregation instead of structural differences between strip cropping and monocultures; with enough data points things tend to get significant despite large variance. This should have been elaborated further upon by the authors with additional analyses, designed to find out where differences originate and what it tells about the functioning of the system. Or it should have provided ample reason for cautioning in drawing conclusions about the supposed effectiveness of strip cropping based on these findings.

We believe that this is a misunderstanding of our approach. In the field-level analyses we pooled samples from the same field (i.e. pseudo-replicates were pooled), resulting in a relatively small sample size of 50 samples. We revised the methods section to better explain this (line 318-322). Therefore, the statement “with enough data points things tend to get significant” is not applicable here.

(2) The authors report percentages calculated as relative change of species richness and abundance in strip cropping compared to monocultures after rarefaction. This is in itself correct, however, it can be rather tricky to interpret because the perspective on actual species richness and abundance in the fields and treatments is completely lost; the reported percentages are dimensionless. The authors could have provided the average cumulative number of species and abundance after rarefaction. Also, range and/or standard error would have been useful to provide information as to the scale of differences between treatments. This could provide a new perspective on the magnitude of differences between the two treatments which a dimensionless percentage cannot.

We agree that this would be the preferred approach if we would have had a perfectly balanced dataset. However, this approach is not feasible with our unbalanced design and differences in sampling effort. While we acknowledge the limitation of the interpretation of percentages, it does allow reporting relative changes for each combination of location, year and crop. The number of samples on which the percentages were based were always kept equal (through rarefaction) between the cropping systems (for each combination of location, year and crop), but not among crops, years and location. This approach allowed us to make a better estimation whenever more samples were available, as we did not always have an equal number of samples available between both cropping systems. For example, sometimes we had 2 samples from a strip cropped field and 6 from the monoculture, here we would use rarefaction up to 2 samples (where we would just have a better estimation from the monoculture). In other cases, we had 4 samples in both strip cropped and monoculture fields, and we chose to use rarefaction to 4 samples to get a better estimation altogether. Adding a value for actual richness or abundance to the figures would have distorted these findings, as the variation would be huge (as it would represent the number of ground beetle(s) species

per 2 to 6 pitfall samples). Furthermore, the dimension that reviewer 3 describes would thus be “The number of ground beetle species / individuals per 2 to 6 samples”, not a very informative unit either.

(3) The authors appear to not have modelled the abundance of any of the dominant ground beetle species themselves. Therefore it becomes impossible to assess which important species are responsible (if any) for the differences found in activity density between strip cropping and monocultures and the possible life history traits related reasons for the differences, or lack thereof, that are found. A big advantage of using ground beetles is that many life history traits are well studied and these should be used whenever there is reason, as there clearly is in this case. Moreover, it is unclear which species are responsible for the difference in species richness found at the field level. Are these dominant species or singletons? Do the strip cropping fields contain species that are absent in the monoculture fields and are not the cause of random variation or sampling? Unfortunately, the authors do not report on any of these details of the communities that were found, which makes the results much less robust.

Thank you for raising this point. We have reconsidered our indicator species analysis and found that it is rather sensitive for rare species and insensitive to changes in common species. Therefore, we have replaced the indicator species analyses with a GLM analysis for the 12 most common genera of ground beetles in the revised manuscript. This will allow us to go more in depth on specific traits of the genera which abundances change depending on the cropping system. In the revised manuscript, we will also discuss these common genera more in depth, rather than focusing on rarer species (line 135-143, 182-200 in discussion). Furthermore, we have added information on rarity and habitat preference to the table that shows species abundances per location (Table S2), and mention these aspects briefly in the results (line 145-153).

(4) In the discussion they conclude that there is only a limited amount of interstrip movement by ground beetles. Otherwise, the results of the crop-level statistical tests would have shown significant deviation from corresponding monocultures. This is a clear indication that the strips function more like mini-monocultures instead of being more than the sum of its parts.

This is in line with our point in the first paragraph of the discussion and an important message of our manuscript.

(5) The RDA results show a modelled variable of differences in community composition between strip cropping and monoculture. Percentages of explained variation of the first RDA axis are extremely low, and even then, the effect of location and/or year appear to peak through (Figure S3), even though these are not part of the modelling. Moreover, there is no indication of clustering of strip cropping on the RDA axis, or in fact on the first principal component axis in the larger RDA models. This means the explanatory power of different treatments is also extremely low. The crop level RDA's show some clustering, but hardly any consistent pattern in either communities of crops or species correlations, indicating that differences between strip cropping and monocultures are very small.

We agree and we make a similar point in the first paragraph of the discussion (line 160-162).

Furthermore, there are a number of additional weaknesses in the paper that should be addressed:

The introduction lacks focus on the issues at hand. Too much space is taken up by facts on insect decline and land sharing vs. land sparing and not enough attention is spent on

the scientific discussion underlying the statements made about crop diversification as a restoration strategy. They are simply stated as facts or as hypotheses with many references that are not mentioned or linked to in the text. An explicit link to the results found in the large number of references should be provided.

We revised the introduction by omitting the land sharing vs. land sparing topic and better linking references to our research findings.

The mechanistic understanding of strip cropping is what is at stake here. Does strip cropping behave similarly to intercropping, a technique that has been proven to be beneficial to biodiversity because of added effects due to increased resource efficiency and greater plant species richness? This should be the main testing point and agenda of strip cropping. Do the biodiversity benefits that have been shown for intercropping also work in strip cropping fields? The ground beetles are one way to test this. Hypotheses should originate from this and should be stated clearly and mechanistically.

We agree with the reviewer and clarified this research direction clearer in the introduction of the revised manuscript (line 66-72).

One could question how useful indicator species analysis (ISA) is for a study in which predominantly highly eurytopic species are found. These are by definition uncritical of their habitat. Is there any mechanistic hypothesis underlying a suspected difference to be found in preferences for either strip cropping or monocultures of the species that were expected to be caught? In other words, did the authors have any a priori reasons to suspect differences, or has this been an exploratory exercise from which unexplained significant results should be used with great caution?

Point well taken. We agree that the indicator species analysis has limitations and therefore now replaced this with GLM analysis for the 12 most common ground beetle genera.

However, setting these objections aside there are in fact significant results with strong species associations both with monocultures and strip cropping. Unfortunately, the authors do not dig deeper into the patterns found a posteriori either. Why would some species associate so strongly with strip cropping? Do these species show a pattern of pitfall catches that deviate from other species, in that they are found in a wide range of strips with different crops in one strip cropping field and therefore may benefit from an increased abundance of food or shelter? Also, why would so many species associate with monocultures? Is this in any way logical? Could it be an artifact of the data instead of a meaningful pattern? Unfortunately, the authors do not progress along these lines in the methods and discussion at all.

We thank reviewer 3 for these valuable perspectives. In the revised manuscript, we further explored the species/genera that respond to cropping systems and discuss these findings in more detail in the revised manuscript (line 182-200 in discussion).

A second question raised in the introduction is whether the arable fields that form part of this study contain rare species. Unfortunately, the authors do not elaborate further on this. Do they expect rare species to be more prevalent in the strip cropping fields? Why? Has it been shown elsewhere that intercropping provides room for additional rare species?

The answer is simply no, we did not find more rare species in strip cropping. In the revised manuscript, we added a column for rarity (according to waarneming.nl) in the table showing abundances of species per location (table S2). We only found two rare species, one of which

we only found a single individual and one that was more related to the open habitat created by a failed wheat field. We discuss this more in depth in the revised results (line 145-153).

Considering the implications the results of this research can have on the wider discussion of bending the curve and the effects of agroecological measures, bold claims should be made with extreme restraint and be based on extensive proof and robust findings. I am not convinced by the evidence provided in this article that the claim made by the authors that strip cropping is a useful tool for bending the curve of biodiversity loss is warranted.

We believe that strip cropping can be a useful tool because farmers readily adopt it and it can result in modest biodiversity gains without yield loss. However, strip cropping is indeed not a silver bullet (which we also don't claim). We nuanced the implications of our study in the revised manuscript (line 30-35, 232-237).

Reviewer #3 (Recommendations for the authors):

General comments:

(1) I am missing the R script and data files in the manuscript. This is a serious drawback in assessing the quality of the work.

Datasets and R scripts will be made available upon completion of the manuscript.

(2) I have doubts about the clarity of the title. It more or less states that strip cropping is designed in order to maintain productivity. However, the main objective of strip cropping is to achieve ecological goals without losing productivity. I suggest a rethink of the title and what it is the authors want to convey.

As the title lead to false expectations for multiple reviewers regarding analyses on yield, we chose to alter the title and removed any mention of yield in the title.

(3) Line 22: I would add something along the lines of: "As an alternative to intercropping, strip cropping is pioneered by Dutch farmers..." This makes the distinction and the connection between the two more clear.

In our opinion, strip cropping is a form of intercropping. We have changed this sentence to reflect this point better. (line 21-22)

(4) Line 24: "these" should read "they"

After changing this sentence, this typo is no longer there (line 24).

(5) Line 34-48. I think this introduction is too long. The paper is not directly about insect decline, so the authors could consider starting with line 43 and summarising 34-42 in one or two sentences.

Removed a sentence on insect declines here to make the introduction more streamlined.

(6) Line 51-59. I am not convinced the land sparing - land sharing idea adds anything to the paper. It is not used in the discussion and solicits much discussion in and of itself unnecessary in this paper. The point the authors want to make is not arable fields compared to natural biodiversity, but with increases in biodiversity in an already heavily degraded ecosystem; intensive agriculture. I think the introduction should focus on that narrative, instead of the land sparing-sharing dichotomy, especially because too little attention is spent on this narrative.

We removed the section on land-sparing vs land-sharing as it was indeed off-topic.

(7) Line 85. *Dynamics is not correctly used here. It should read Ground beetle communities are sensitive.*

Changed accordingly (line 78-79).

(8) Line 90-91. *Here, it should be added that ground beetles are used as indicators for ground-dwelling insect diversity, not wider insect diversity in agricultural systems. In fact, Gerlach et al., the reference included, clearly warn against using indicator groups in a context that is too wide for a single indicator group to cover and Van Klink (2022) has recently shown in a meta-analysis that the correlation between trends in insect groups is often rather poor.*

We removed the sentence that claimed ground beetles to be indicators of general biodiversity, and have focused the text in general more on ground beetle biodiversity, rather than general biodiversity.

(9) Line 178: *was there a high weed abundance measured in the stripcropping fields? Or has there been reports on higher weed abundance in general? The references provided do not appear to support this claim.*

To our knowledge, there is only one paper on the effect of strip cropping on weeds (Ditzler et al., 2023). This paper shows strip cropping (and more diverse cropping systems) reduce weed cover, but increase weed richness and diversity. We mistakenly mentioned that crop diversification increases weed seed biomass, but have changed this accordingly to weed seed richness. The paper from Carbonne et al. (2022) indeed doesn't show an effect of crop diversification on weeds. However, it does show a positive relation between weed seed richness and ground beetle activity density. We have moved this citation to the right place in the sentence (line 172-175).

(10) Line 279-288. *The description of sampling with pitfalls is inadequate. Please follow the guidelines for properly incorporating sufficient detail on pitfall sampling protocols as described in Brown & Matthews 2016,*

We were sadly not aware of this paper prior to the experiments, but have at least added information on all characteristics of the pitfall traps as mentioned in the paper (line 290-294).

(11) Lines 307-310. *What reasoning lies behind the choice to focus on the most beetle-rich monocultures? Do the authors have references for this way of comparing treatments? Is there much variation in the monocultures that solicits this approach? It would be preferable if the authors could elaborate on why this method is used, provide references that it is a generally accepted statistical technique and provide additional assessments of the variation in the data so it can be properly related to more familiar exploratory data analysis techniques.*

We ran two analyses for the field-level richness and abundance. First we used all combinations of monocultures and strip cropping. However, as strip cropping is made up of (at least) 2 crops, we had 2 constituent monocultures. As we would count a comparison with the same strip cropped field twice when we included both monocultures, we also chose to run the analyses again with only those monocultures that had the highest richness and abundance. This choice was done to get a conservative estimate of ground beetle richness

increases through strip cropping. We explained this methodology further in the statistical analysis section (line 329-335).

In Figure S6 the order of crop combinations is altered between 2021 on the left and 2022 on the right. This is not helpful to discover any possible patterns.

We originally chose this order as it represented also the crop rotations, but it is indeed not helpful without that context. Therefore, we chose to change the order to have the same crop combinations within the rows.

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